VS-02 Data Standardization and Integration

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1. Purpose and value created

VS-02 makes data from different sources speak the same language. It takes raw data that is already in NSI custody and brings it into the common structure, format, and classifications of the NSI. It does this without changing any value. Only the representation changes: fields are renamed and restructured, types are cast, source codes are mapped one-to-one to standard codes, and descriptive metadata is attached. Direct identifiers are removed or replaced with pseudonymous keys. The value created is **comparability without loss of content**. Data that looked different at the source now uses the same structure, the same codes, and documented definitions, units, and reference periods. Because no value changes, the trace back to the authentic delivery in State #1 stays simple and deterministic.

VS-02 corresponds to the business domain **Standardization without loss of content** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Data quality analyst or operations staff:** the stages in attribute 3, which describe what happens in each production cycle.
- **Methodologist, classification steward, or data architect:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.

- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

> **i** Signature discipline: standardization without loss of content
>
> VS-02 never changes a value. It re-expresses the structure and format, adds standard codes through deterministic one-to-one mapping, and attaches metadata. Unit and currency conversion, reference-period alignment, reconciliation of differing definitions, derivation, and interpretive coding of free text are not VS-02 activities. They belong to VS-03, where every change is documented as an edit. The removal or pseudonymization of direct identifiers is not a loss of content, because identifiers are personal data, not statistical content.

PLACEHOLDER: ArchiMate view VS-02 (a) value stream and capabilities

2. Position and state transition

VS-02 is the second primary value stream. It sits between acquisition (VS-01) and editing (VS-03). It receives Raw Data when VS-01 publishes the *Raw Data Registered* event, and it hands over Standardized Data to VS-03.

VS-02 takes data from State #1 to State #2. Before VS-02, the data is a registered raw version in its original form, with direct identifiers still present. After VS-02, it is a registered, immutable version in the canonical target data model, with standard codes stored next to the original source values, with recorded units and reference periods, and without direct identifiers. The quality gate, the minimum metadata set, and the protection of State #2 are described in *State #2: Standardized Data* and are not repeated here.

State #2 is optional. A source that already arrives in the NSI's standard structure and coding can go directly from #1 to #3. In that case the NSI records in the catalog that #2 was skipped, and why (see *Applying the steady states pragmatically*). Where #2 is skipped, the removal or pseudonymization of direct identifiers must still take place before #3, and it must still be recorded. [**VC12**]

From	To	Gate	Event out
#1 Raw Data	#2 Standardized Data	Gate of State #2 (structural conformance, classification conformance, mapping coverage, semantic metadata, no loss of content, identifier protection, referential integrity, metadata and lineage)	*Standardized Data Registered*

PLACEHOLDER: ArchiMate view VS-02 (c) state handover and events: Raw Data (#1) in the upstream domain, the standardization service at the boundary (profiling, one-to-one code mapping, semantic alignment, conformance check), Standardized Data (#2) inside the standardization domain

3. Stages

The stages describe the value that VS-02 adds in every production cycle. Together they form the execution process type of VS-02. In a mature steady state, the target model, the correspondence tables, and the conformance rules have been designed and built in advance (see attribute 4). Execution then focuses on

applying them, capturing lineage and conformance metadata, and passing the quality gate without changing any value.

Stage	Name	Value added
VS-02.1	Profile and Assess Data Structure	It is clear how far the raw data is from the target standard
VS-02.2	Apply Standard Classifications and Formats	Source codes are expressed in standard classifications and a common format; no value is changed
VS-02.3	Align Metadata and Semantics	The same concept means the same thing across datasets
VS-02.4	Validate Structural and Semantic Conformance	The data conforms to the standards, and no content was lost
VS-02.5	Register Standardized Artifacts	Standardized data is registered and available for reuse (State #2)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-02.1 Profile and Assess Data Structure

Value created: *A clear view of how far the incoming raw data is from the target standard, so that the right mapping and format coercion can be applied.*

The stage starts when VS-01 publishes the *Raw Data Registered* event. The profiling components analyze the structure of the incoming version. They infer types, measure cardinalities and value distributions, and detect missing fields and structural anomalies. The result is compared with the target data model. This produces a structure-gap assessment: which fields map directly, which need format coercion, which source codes have a standard correspondence, and which values are uncoded and must go to VS-03. A significant or unexpected change in structure, such as a source that has changed its schema, goes back to Design. It is not solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Profile the incoming raw data: types, cardinalities, distributions, anomalies	Data profiling report	PRD-5.1 Integrate data	1.3.1 Data standardization
Compare the structure with the target data model	Structure-gap assessment	PRD-5.1 Integrate data	
Identify the format coercions and code mappings to apply	Mapping plan for the dataset	PRD-5.1 Integrate data; PRD-5.2 Classify and code	
Flag unexpected structural deviations for design review	Schema-change exception (if any)	PRD-5.1 Integrate data	

VS-02.2 Apply Standard Classifications and Formats

Value created: *Source codes re-expressed in shared, authoritative classifications and in a common format, so that data is comparable across sources, with no value changed.*

This is the core stage of VS-02. The mapping and format-coercion component applies the designed value-preserving operations: renaming, restructuring, type casting, reformatting, and key alignment. The mapping services then translate values that already carry a source code into the standard classification, such as NACE, ISCO, COICOP, NUTS, or SDMX code lists. The translation uses maintained correspondence tables, one-to-one, and the original value is kept next to the standard code. Values without a correspondence, and free text that needs interpretation, are not coded here. They are flagged and listed for VS-03. Every applied mapping and classification version is recorded as lineage metadata, in line with SAF-P09.

Activity	Key outputs	SAF process	Other reference
Apply format coercion: rename, restructure, cast, reformat, align keys	Format-coerced dataset (values unchanged)	PRD-5.1 Integrate data	1.3.1 Data standardization
Map source codes to standard classifications one-to-one through correspondence tables	Standard-coded dataset (original values kept)	PRD-5.2 Classify and code; DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Flag unmapped values and free text, and list them for VS-03 Data Editing and Enrichment	List of unmapped values	PRD-5.2 Classify and code	
Record the applied mappings and classification versions as lineage	Mapping and classification lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

PLACEHOLDER: VS-02 classification mapping: survey, administrative, and emerging-source inputs passing format coercion and one-to-one correspondence tables; values with a source-code correspondence become standard-coded values with the original kept; unmapped values and free text are routed to VS-03 for interpretive coding

VS-02.3 Align Metadata and Semantics

Value created: *Coherent, shared meaning across datasets, so that the same concept means the same thing everywhere.*

Once structure and codes are standardized, the descriptive metadata is harmonized. This happens at the level of metadata only; no value changes. Each source variable is linked to its canonical concept in the concept system. The unit of measure, the currency, and the reference-period basis of each variable are recorded as received. They are not converted. Where two sources define the same concept differently, the difference is recorded as metadata and flagged for VS-03, which decides how to reconcile it. Quality descriptors, such as completeness and confidentiality flags, are standardized and attached. Downstream consumers then receive both the standardized data and the meaning they need to interpret it.

Activity	Key outputs	SAF process	Other reference
Link source variables to canonical concepts	Concept-aligned variables	OVA-2 Metadata Management	CSDA Metadata Management
Record unit, currency, and reference-period basis per variable (no conversion)	Unit and period metadata	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

Activity	Key outputs	SAF process	Other reference
Record differing cross-source definitions and flag them for VS-03 Data Editing and Enrichment	Definition-difference record	OVA-2 Metadata Management	2.1.4 Statistical register management
Standardize and attach quality descriptors	Harmonized quality metadata	OVA-2 Metadata Management; OVA-1 Quality Management	CSDA Metadata Management

VS-02.4 Validate Structural and Semantic Conformance

Value created: *Confidence that the standardized data conforms to the standards and that no content was lost.*

Before standardized data can be accepted, it must pass the quality gate of State #2. The automated checks from *Design Conformance Validation Rules* run on every dataset. Structural validation checks types, formats, ranges, cardinalities, and referential integrity against the target model. Classification conformance checks that every mapped code exists in the stated classification version, that there are no orphan codes, and that mapping coverage is above the threshold. Semantic conformance checks that units, definitions, and reference periods are recorded for every variable. The content-preservation checks confirm that record counts and key measures are identical to the #1 version, and that no value has changed. Identifier protection is checked against the privacy design.

Every result, exception, and decision is logged in an audit trail. The trail captures data metadata (what was checked, what passed, what failed) and process metadata (who checked, when, which rule version and which engine version). The outcome is one of two business events. *Standardization Conformed* starts registration. *Standardization Non-Conformant* starts exception handling according to the designed rules: re-map, correct the correspondence table, or accept conditionally with a conformance flag. Conditional acceptance is a normal outcome for many sources, because unmapped codes and unmatched keys often have a known cause, such as the lag in registering new businesses.

Activity	Key outputs	SAF process	Other reference
Validate the structure against the target model, including referential integrity	Structural validation result	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	
Validate classification conformance: valid codes, correct version, coverage	Classification conformance result	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	
Validate semantic conformance: units, definitions, reference periods	Semantic conformance result	DLM-2 Promote data through a quality gate	
Run the no-loss-of-content checks: record counts and key measures	Content-preservation result	DLM-2 Promote data through a quality gate	GAMSO Quality Management
Check identifier protection against the privacy design	Identifier protection result	DLM-4 Classify and protect data	SAF-P11 Privacy by design
Create the audit trail: log all results, exceptions, and decisions	Audit trail record	OVA-4 Process Data Management	SAF-P10 Security by design

> **!** Boundary with VS-03
>
> No value changes in VS-02. The following activities belong to VS-03 Data Editing and Enrichment: content correction, imputation, enrichment, unit and currency conversion, reference-period alignment, reconciliation of values across sources, derivation of new variables, and interpretive coding of free text.

VS-02.5 Register Standardized Artifacts

Value created: *Secure, organized, and traceable storage of standardized data that downstream consumers can find.*

This stage persists conformant standardized data and makes it findable for the rest of the statistical system. Each dataset version receives a unique persistent identifier (URI or UUID). It is stored in the standardized data repository, in line with the loose coupling of SAF-P07. The removal or pseudonymization of direct identifiers is confirmed and recorded in the pseudonymization record, without ever recording the keys themselves. Metadata is registered in the data catalog, in line with SAF-P09: mapping lineage, classification versions, concept links, recorded units and periods, the list of unmapped values, and the conformance assessment. Field-level lineage from State #1 to State #2 is recorded, so every standardized value can be traced to the raw value and the mapping rule. Access permissions are applied according to the privacy design. A rerun with new rules, such as a new correspondence table, always creates a new version; it never overwrites an earlier one. The stage ends by publishing the *Standardized Data Registered* event, which tells VS-03 that new standardized data is available.

Output: Standardized Data (State #2), the official handover artifact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #2: Standardized Data*.

Activity	Key outputs	SAF process	Other reference
Assign a unique persistent identifier (URI or UUID) to the dataset version	Dataset ID	DLM-3 Register and catalog data	
Store the data in the standardized data repository	Stored standardized artifact	DLM-5 Store and version data	SAF-P07 Implement loosely coupled processes and systems
Confirm and record identifier removal or pseudonymization	Pseudonymization record	DLM-4 Classify and protect data	SAF-P11 Privacy by design
Register metadata in the data catalog: mapping lineage, classifications, concepts, units and periods, unmapped values, conformance	Catalog registration	DLM-3 Register and catalog data; OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Apply access permissions according to the privacy design	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10 Security by design, SAF-P11 Privacy by design
Publish the *Standardized Data Registered* event to notify VS-03 Data Editing and Enrichment	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-02 standardized data artifact structure: the standardized dataset version linked to its mapping lineage, classification record, conformance assessment, and pseudonymization and access record

4. The four process types

VS-02 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-02. Their names describe the work in the context of standardization. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-02 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-02 these are above all the target data model, the correspondence tables, and the conformance rules. Without them, execution produces data that cannot be compared, and management has no baseline to measure against. Design runs when a new product or source is introduced, when a standard classification is revised, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Target Data Models and Schemas	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data	Target data model, canonical concept catalog, structural conformance rules, integration key design
Design Classification Mapping Strategy	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Classification standards register, version policy, correspondence tables, routing rules for unmapped values
Design Metadata and Semantic Harmonization Rules	PRD-2.2 Design variables; DLM-1 Plan and design data assets; OVA-2 Metadata Management	Harmonized concept system, unit and period metadata standards, semantic mapping rules, definition-difference policy
Design Conformance Validation Rules	PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management; OVA-2 Metadata Management	Structural, classification, and semantic conformance rules, content-preservation checks, exception handling policy
Design Standardization Registration and Privacy Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management	Data classification, access control design, lineage model, pseudonymization design, audit logging specification

Design Target Data Models and Schemas

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data.

Standardization starts with the canonical structure that all data must conform to. This sub-process specifies the target data model of Standardized Data (State #2): entities, fields, data types, formats, cardinalities, and referential constraints. It sets canonical representations for common concepts, such as dates, geographies, currencies, and identifiers of persons and businesses. Data from surveys, registers, and external sources can then be expressed in the same way. The team also defines the structural conformance contract, which states when a dataset is structurally valid against the target model. Where several sources describe the same population, the design defines a common key structure, such as the key of the statistical business register. The actual linkage of records takes place in VS-03; VS-02 only aligns the keys.

Activity	Key outputs	SAF process	Other reference
Define the target data model of Standardized Data (State #2): entities, fields, types, formats	Target data model specification	PRD-2.2 Design variables; DLM-1 Plan and design data assets	1.3.1 Data standardization
Define canonical representations for common concepts: dates, geography, currency, identifiers	Canonical concept catalog	PRD-2.2 Design variables; DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Define the structural conformance contract and referential constraints	Structural conformance rules	PRD-2.5 Design processing and analysis	
Define the key structure for aligning several sources	Integration key design	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	2.1.4 Statistical register management

Design Classification Mapping Strategy

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

Once the target structure is set, the classification steward designs how source codes are mapped to standard classifications. This sub-process selects the authoritative classifications and code lists that the NSI conforms to, such as NACE, ISCO, COICOP, NUTS, and SDMX code lists. It defines the version policy: which version applies, and how a transition between versions is handled. The mapping is deterministic. A maintained correspondence table translates a value that already carries a source code into the matching standard code, and the original value is kept. The design also specifies what happens to values without a correspondence. They are flagged and passed to VS-03, which carries out any interpretive coding of free text. The reference and master data that anchor the mapping are identified, so that the correspondence tables stay aligned with authoritative sources.

Activity	Key outputs	SAF process	Other reference
Select the authoritative classifications and code lists	Classification standards register	PRD-2.2 Design variables; DLM-9 Manage reference and master data	1.3.1 Data standardization
Define the classification version policy and the handling of transitions	Classification version policy	DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Design deterministic one-to-one correspondence tables from source codes to standard codes	Set of correspondence tables	PRD-2.5 Design processing and analysis	
Define the handling of unmapped values and free text: flag and route to VS-03 Data Editing and Enrichment	Routing rules for unmapped values	PRD-2.5 Design processing and analysis	
Identify the reference and master data sources for mapping	Reference data catalog	DLM-9 Manage reference and master data	CSDA Reference Data

i Interpretive coding is not part of VS-02

Coding a written job title to ISCO, or a free-text product name to COICOP, requires judgment. That judgment can change the informational content, so it needs review and an edit record. It is therefore carried out in VS-03. VS-02 only applies deterministic correspondences from a source code to a standard code.

Design Metadata and Semantic Harmonization Rules

Refines: PRD-2.2 Design variables; DLM-1 Plan and design data assets; OVA-2 Metadata Management.

This sub-process defines how descriptive metadata is harmonized, so that data from different sources can be understood in the same way. No value is changed by these rules. The team designs the harmonized definitions of concepts and variables. It defines the metadata that records the unit of measure, the currency, and the reference-period basis of each variable; these are recorded, not converted. Semantic mapping rules link source variables to canonical concepts in the concept system. Where sources use different definitions for the same concept, the design states that the difference is recorded as metadata and flagged for VS-03. Finally, the catalog structure for standardized artifacts is designed so that lineage from State #1 to State #2 can be fully traced. This makes active metadata a foundation for automation downstream, as emphasized by the Metadata Management capability of CSDA.

Activity	Key outputs	SAF process	Other reference
Define harmonized definitions of concepts and variables	Harmonized concept system	PRD-2.2 Design variables; OVA-2 Metadata Management	2.1.2 Metadata management
Define the unit, currency, and reference-period metadata to record per variable (no conversion)	Unit and period metadata standards	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Design semantic mapping rules from source variables to canonical concepts	Semantic mapping rules	OVA-2 Metadata Management	CSDA Metadata Management
Design the rules to record differing definitions as metadata and flag them for VS-03 Data Editing and Enrichment	Definition-difference policy	OVA-2 Metadata Management	
Design the catalog structure and the lineage model for standardized artifacts	Catalog registration template	DLM-1 Plan and design data assets; OVA-2 Metadata Management	CSDA Metadata Management

Design Conformance Validation Rules

Refines: PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management; OVA-2 Metadata Management.

Before any data is standardized, it must be clear when the result counts as standardized. This sub-process sets the structural validation rules: field types, formats, value ranges, cardinalities, and referential integrity against the target model. It sets the classification conformance rules: every mapped code belongs to the correct classification version, and there are no orphan codes. It sets the semantic conformance rules: units, definitions, and reference periods are recorded for every variable. The coverage threshold is agreed, which is the maximum acceptable share of unmapped values. Content-preservation checks verify that record counts and key measures are identical before and after standardization, and that no value has changed. Exception handling rules distinguish between rejection, conditional acceptance with a conformance flag, and routing

of unmapped values to VS-03. Together, these rules make up the VS-02 part of the quality gate of State #2.

Activity	Key outputs	SAF process	Other reference
Define structural validation rules against the target model	Structural validation rule set	PRD-2.5 Design processing and analysis	
Define classification conformance rules: valid codes, correct version, no orphans	Classification conformance rules	PRD-2.5 Design processing and analysis	
Define semantic conformance rules: units, definitions, reference periods	Semantic conformance rules	PRD-2.5 Design processing and analysis; OVA-2 Metadata Management	
Define the coverage threshold and the content-preservation checks	Content-preservation checks, coverage threshold	DLM-2 Promote data through a quality gate; OVA-1 Quality Management	GAMSO Quality Management
Design exception handling rules: rejection, conditional acceptance, routing to VS-03 Data Editing and Enrichment	Exception handling policy	DLM-2 Promote data through a quality gate	

Design Standardization Registration and Privacy Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management.

This last design sub-process makes sure that standardized artifacts are registered correctly, can be traced, and comply with privacy rules. Each standardized dataset receives a classification for availability, integrity, and confidentiality, in line with SAF-P09. Role-based access controls for the standardized data repository are designed in line with SAF-P10. The lineage model makes sure that every standardized field can be traced to its raw value and to the mapping rule that was applied. Privacy protection follows SAF-P11. State #2 is the point where direct identifiers leave the working data. The design therefore decides per identifier whether it is removed or pseudonymized, which method is used, and how the keys are managed separately from the data. Audit logging requirements, which define the mapping and coercion events that must be recorded, complete the design.

Activity	Key outputs	SAF process	Other reference
Define the data classification (availability, integrity, confidentiality) for standardized datasets	Data classification policy	DLM-4 Classify and protect data	SAF-P09 No data without metadata and classification
Design role-based access control for the standardized data repository	Access control design	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control
Design the lineage model: standardized field, raw value, mapping rule	Lineage model specification	DLM-1 Plan and design data assets	CSDA Traceability

Activity	Key outputs	SAF process	Other reference
Design identifier removal and pseudonymization: which identifiers, method, key management, minimization	Pseudonymization design	DLM-4 Classify and protect data	SAF-P11 Privacy by design; 3.3.1 Privacy management
Design audit logging requirements for mapping and coercion	Audit logging specification	OVA-4 Process Data Management	SAF-P10 Security by design

PLACEHOLDER: VS-02 design process decomposition: inputs (State #1 schema, classification standards, framework standards, legal and privacy requirements), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the standardization pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Classification and mapping tooling in particular is a good candidate for sharing between NSIs.

Implementation sub-process	Refines	Main outputs
Build Profiling and Mapping Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Profiling module, configured mapping and format-coercion component, versioned mapping repository
Configure Classification Mapping Services	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Configured mapping services, version-controlled classifications, routing of unmapped values
Build Conformance Validation Pipelines and Metadata Registration	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate; DLM-3 Register and catalog data; DLM-4 Classify and protect data; OVA-2 Metadata Management	Automated conformance pipeline, pseudonymization component, metadata propagation, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, mapping verification, production-ready pipeline, operational procedures

Build Profiling and Mapping Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the target data model and the mapping designs into operational tools. Profiling components are built to analyze the structure of incoming raw data. They infer types and detect cardinalities, value distributions, and structural anomalies, so the pipeline can report how far each source is from the target model. The mapping and format-coercion component is built or configured to apply the designed value-preserving operations: renaming, restructuring, type casting, reformatting, and key alignment. It

does not convert units, currencies, or reference periods. The component is declarative and records, for each output field, which input field and which rule produced it. All mapping logic is version-controlled, so that any standardized record can be reproduced from its raw version and the recorded rule versions.

Activity	Key outputs	SAF process	Other reference
Build data profiling components: type inference, cardinality, distributions, anomalies	Production-ready profiling module	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Build or configure the mapping and format-coercion component	Configured mapping component	PRD-3.2 Reuse or build processing and analysis components	1.3.1 Data standardization
Configure lineage capture for every mapping and coercion operation	Lineage-aware mapping modules	PRD-3.4 Configure workflows	CSDA Traceability
Put all mapping and coercion logic under version control	Versioned mapping repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Configure Classification Mapping Services

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

In this sub-process the designed correspondence tables become automated mapping services. The correspondence tables are loaded and connected to the authoritative reference data. A value that already carries a source code is then translated one-to-one into the matching standard code. Version control of classifications is configured, so that the correct version is applied and transitions between versions are managed. Values without a matching correspondence, and free-text values, are routed to the list for VS-03 instead of being coded here. Connections to reference and master data are configured, so that the mapping stays aligned with the authoritative registers.

Activity	Key outputs	SAF process	Other reference
Load the correspondence tables for deterministic mapping from source codes to standard codes	Configured mapping services	PRD-3.2 Reuse or build processing and analysis components; DLM-9 Manage reference and master data	
Configure version control of classifications and the handling of transitions	Version-controlled classifications	DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Configure the routing of unmapped values and free text to VS-03 Data Editing and Enrichment	Routing of unmapped values	PRD-3.4 Configure workflows	
Configure connections to reference and master data	Synchronized reference data	DLM-9 Manage reference and master data	CSDA Reference Data

Build Conformance Validation Pipelines and Metadata Registration

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate; DLM-3 Register and catalog data; DLM-4 Classify and protect data; OVA-2 Metadata Management.

This sub-process implements the conformance rules and the registration of standardized artifacts. The structural, classification, semantic, and content-preservation checks are implemented as automated validations that run on every standardized dataset. The removal or pseudonymization of identifiers is implemented according to the privacy design. Catalog entries are built from the designed metadata model. Automatic propagation is configured, so that mapping lineage, classification versions, concept links, and conformance scores flow into the catalog without manual work. Event notifications are configured, so that VS-03 is alerted when standardized data is registered. Exception workflows route non-conforming data to the right queue.

Activity	Key outputs	SAF process	Other reference
Implement the conformance checks as automated validations	Automated conformance pipeline	PRD-3.2 Reuse or build processing and analysis components; DLM-2 Promote data through a quality gate	
Implement identifier removal and pseudonymization according to the privacy design	Pseudonymization component	PRD-3.2 Reuse or build processing and analysis components; DLM-4 Classify and protect data	SAF-P11 Privacy by design
Configure automatic propagation of metadata and lineage into the catalog	Automated metadata pipelines	PRD-3.4 Configure workflows; DLM-3 Register and catalog data; OVA-2 Metadata Management	CSDA Metadata Management
Configure the *Standardized Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows	SAF-P07 Implement loosely coupled processes and systems
Configure exception handling workflows	Routing of non-conforming data, review queues	PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate	

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all individual components: profilers, the mapping component, mapping services, validators, and the catalog. Integration testing checks the end-to-end flow from the *Raw Data Registered* event through mapping and validation to catalog registration. A pilot with real or synthetic data checks that source codes are mapped correctly, that lineage is complete, and that conformance criteria are enforced. The pilot also confirms that no value changes, that unmapped values are routed to VS-03, and that identifier removal works as designed. The correspondence tables are verified against the source code lists. Performance testing confirms that the pipeline handles the expected volumes within the processing window. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	

Activity	Key outputs	SAF process	Other reference
Test integration: end-to-end flow from raw intake to registration	Integration test report	PRD-3.5 Test production systems	
Run a pilot with real or synthetic data: lineage, no loss of content, routing, identifier removal	Pilot report, validated lineage	PRD-3.6 Test statistical business process	SAF-P11 Privacy by design
Verify the correspondence tables against the source code lists	Mapping verification report	PRD-3.6 Test statistical business process	
Test performance against expected volumes	Performance test report	PRD-3.5 Test production systems	
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-02 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, pilot and mapping verification, and performance and lineage testing, resulting in a production-ready standardization pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-02 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-01 publishes the *Raw Data Registered* event, or when a rule changes and an existing #1 version must be standardized again. The stages then run in order for each raw version. Execution applies the designs and the solution as they are; a new source structure or a missing correspondence goes back to Design. Every run ends in one of three outcomes: registered, registered with conformance flags, or returned for re-mapping.

Aspect	VS-02
Trigger	*Raw Data Registered*; a new version of a correspondence table, classification, or target model
Cadence	Per registered raw version
Performed by	ROL-14 Data quality analyst, ROL-03 Data steward, ROL-09 Classification steward
Input	Raw data version (#1), target data model, correspondence tables, privacy design
Output	Registered standardized data version (#2), its conformance assessment, and the list of unmapped values for VS-03 Data Editing and Enrichment
Control point	Gate of State #2 in VS-02.4 Validate Structural and Semantic Conformance
Exception path	Re-mapping or correction of a correspondence table; structural change at the source back to Design; failed identifier protection escalated to ROL-19 Compliance officer [**VC13**]

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the standardization pipeline, assesses the conformance and classification quality of standardized data, makes sure the domain complies with the rules, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Standardization Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Mapping completeness report, conformance quality report, trend analysis
Manage Classification and Metadata Lifecycle	DLM-9 Manage reference and master data; OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use	Classification version and correspondence register, metadata quality report, lineage audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Audit trail summary, privacy and standards compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-02 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the processing window. They track paradata: who performed each step, when, how long it took, which engine and classification versions were used, and which exceptions occurred. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO for monitoring performance across the production cycle.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the processing window	Throughput report	OVA-4 Process Data Management	
Track paradata: operator, timing, engine and classification versions, exceptions	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Standardization Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the standardized data itself is evaluated against the criteria from *Design Conformance Validation Rules*. Mapping completeness covers the share of source codes mapped

one-to-one and the share of values routed to VS-03. Classification conformance covers orphan codes and the use of the correct version. Structural conformance covers alignment with the target model. Metadata completeness covers whether units, reference periods, and definitions are recorded. Content preservation confirms that no value changed and that record counts and key measures are identical. Quality indicators are produced per dataset, per source, and per production cycle. Trends are tracked to detect deteriorating sources, growing shares of unmapped values, or problems with classification versions. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Assess mapping completeness and the share of unmapped values	Mapping completeness report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess classification conformance and the rate of orphan codes	Classification conformance report	OVA-1 Quality Management	CSDA Data Quality
Assess structural conformance, metadata completeness, and content preservation	Conformance quality report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across sources, classifications, and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Classification and Metadata Lifecycle

Refines: DLM-9 Manage reference and master data; OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use.

The classifications and metadata that support standardization must stay complete, accurate, and current. This sub-process governs the life cycle of classification versions. It adopts new versions, such as a revision of NACE or COICOP, manages the transition period, and maintains the correspondence tables between versions. It verifies that every standardized dataset carries its full mapping lineage, classification versions, and concept links, as a compliance check on SAF-P09. The concept system, code lists, and reference data are maintained, so that the mapping stays anchored to authoritative sources. Lineage from State #1 to State #2 is tracked. Reconciliation across systems makes sure that catalog entries match the artifacts actually stored.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle of classification versions and correspondence tables	Classification version and correspondence register	DLM-9 Manage reference and master data	SAF-P09 No data without metadata and classification
Verify metadata completeness for all standardized datasets	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Maintain the concept system, code lists, and reference data	Reference data maintenance log	DLM-9 Manage reference and master data; OVA-2 Metadata Management	CSDA Metadata Management
Track metadata lineage from raw to standardized data	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability
Reconcile metadata across systems	Catalog-artifact reconciliation	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-02 operates within legal, regulatory, and organizational requirements. Audit trail reports show the full traceability of every standardized value, from the raw value through the mapping rules that were applied. This supports reproducibility and proves that no content was lost. Privacy compliance reporting confirms that direct identifiers were removed or pseudonymized at the transition from State #1 to State #2, and that data minimization was applied. Standards compliance reporting confirms conformance to the adopted classifications and exchange standards, such as SDMX. Security compliance reporting verifies that the designed access controls and audit logging were applied. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce audit trail reports: full lineage from raw value to standardized value	Audit trail summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on privacy compliance: identifier removal or pseudonymization, minimization	Privacy compliance record	OVA-1 Quality Management	SAF-P11 Privacy by design; 3.3.1 Privacy management
Report on standards compliance: classifications, SDMX exchange	Standards compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: measures applied, breaches, incidents	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: is mapping coverage rising, is the share of unmapped values falling, and are fewer datasets returned for re-mapping? Improvement opportunities are identified. Examples are extending correspondence tables to raise coverage, refining mappings that produce orphan codes, preparing the adoption of a new classification version, and simplifying conformance rules that produce too many false alarms. A recurring volume of unmapped values from one source is reported to VS-03, because the solution is interpretive coding, not a wider VS-02. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Where a new method is needed, such as machine-learning support for coding, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: extension of correspondence tables, refined mappings, version adoption	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management

Activity	Key outputs	SAF process	Other reference
Feed recommendations back to the design sub-processes, to VS-03 Data Editing and Enrichment, or to VS-08 Innovation and Methodological Development	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-02 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, coding needs to VS-03, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-02, metadata, and classification in particular, is not a by-product but the central asset being created. The table shows what each process type contributes. Data metadata describes the datasets. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Target data models, classification schemes, code lists, correspondence tables, conformance criteria, harmonized definitions	Log of design decisions, classification version policy, review and approval records
Implementation	Mapping configurations, parameters of the mapping services, parameters of validation rules, catalog templates	Build log, test results, records of configuration changes, deployment approvals
Execution	Mapping lineage per dataset, applied classification versions, recorded units and periods, conformance scores, profiling statistics, list of unmapped values	Operator identity, timing, volumes, mapping coverage, number of unmapped values, engine version
Management	Conformance scores, classification quality assessments, coverage reports, lineage assessments, trend analyses	Conformance dashboards, classification quality trends, improvement decisions, audit findings

PLACEHOLDER: VS-02 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns Raw Data (#1) into Standardized Data (#2), generating mapping lineage, conformance, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-02 is an internal value stream. Its only external actor is the body that maintains the classification and exchange standards the NSI conforms to. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the system configurator in the original VS-02 description is ROL-11 Platform engineer. VS-01, which supplies the raw data, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. **[VC14]**

Actor or role	ID	Process type	Responsibility in VS-02 Data Standardization and Integration
International statistical organization	ACT-07	External	Maintains classification and exchange standards, such as NACE, ISCO, COICOP, and SDMX (for example Eurostat and UNSD)
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-02 Data Standardization and Integration
Data owner	ROL-02	Management	Accountable for State #2, for conformance, and for applying the privacy design
Data steward	ROL-03	Execution	Registers standardized datasets, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention; manages pseudonymization keys separately from the data
Methodologist	ROL-07	Design	Designs the mapping method and the harmonization of definitions across sources
Metadata and data architect	ROL-08	Design	Designs the target data model, canonical concepts, semantic mappings, and the metadata schema
Classification steward	ROL-09	Design, Execution	Owns the classification strategy, code lists, and correspondence tables; maintains classification versions
Data engineer	ROL-10	Implementation	Builds and configures the profiling, mapping, and coercion components and the pipeline

Actor or role	ID	Process type	Responsibility in VS-02 Data Standardization and Integration
Platform engineer	ROL-11	Implementation	Configures classification services, connections to reference data, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, conformance, and integration testing of the pipeline
Data quality analyst	ROL-14	Execution	Profiles data, assesses conformance, and logs unmapped values and exceptions
Quality manager	ROL-18	Management	Monitors key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures privacy and standards compliance and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-02 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-02 Data Standardization and Integration
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the target data model and the mapping strategy, and build the standardization pipeline
1.3 Processing	1.3.1 Data standardization	Carry out the standardization: format coercion, one-to-one code mapping, semantic alignment **[VC15]**
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data and metadata through standardization, harmonize semantics, assess conformance, align keys with reference registers
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain mapping components, classification services, and validation pipelines
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Remove or pseudonymize direct identifiers and build security into the pipeline
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the classification standards, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-02 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. **[VC25]**

Stage	Realized by processes	Main functions
VS-02.1 Profile and Assess Data Structure	PRD-5.3 Review and validate	FUN-3.3 Process statistical data
VS-02.2 Apply Standard Classifications and Formats	PRD-5.2 Classify and code; DLM-9 Manage reference and master data	FUN-3.3 Process statistical data; FUN-7.4 Manage reference and master data
VS-02.3 Align Metadata and Semantics	OVA-2 Metadata Management	FUN-7.2 Manage metadata
VS-02.4 Validate Structural and Semantic Conformance	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	FUN-3.3 Process statistical data; FUN-7.3 Manage data lifecycle and storage
VS-02.5 Register Standardized Artifacts	DLM-3 Register and catalog data; DLM-4 Classify and protect data; DLM-5 Store and version data	FUN-7.2 Manage metadata; FUN-7.3 Manage data lifecycle and storage; FUN-7.5 Protect data

VS-02 carries out only part of PRD-5.1, PRD-5.2, and PRD-5.3: key alignment without record linkage, deterministic coding without interpretive coding, and structural checks without checks on the correctness of values. The remaining parts of these processes take place in VS-03. **[VC16]**

PLACEHOLDER: ArchiMate view VS-02 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-02 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.2, PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-2, OVA-4	Phase 2 Design (2.2, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Reference and Master Data	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.4 Statistical register management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-2, DLM-3, DLM-4, DLM-9, OVA-2	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Integration (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Execution	PRD-5.1 to PRD-5.3 (partly), DLM-2 to DLM-5, DLM-9, OVA-1, OVA-2, OVA-4	Phase 5 Process (5.1 key alignment only, 5.2 deterministic coding only, 5.3 structural checks only)	Production (execution)	Data Integration (execution)	1.3.1 Data standardization, 2.1.2 Metadata management, 2.1.4 Statistical register management
Management	OVA-1, OVA-2, OVA-4, DLM-8, DLM-9, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-02 most directly.

Principle	Statement	Application in VS-02 Data Standardization and Integration
SAF-P01	Reuse before open source, open source before buying, buying before building	Mapping, coding, and SDMX tooling is reused across products first. Where new tooling is needed, open-source solutions, preferably developed with other NSIs, come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Standardization is independent from acquisition and editing. Standardized Data (State #2) is the decoupling interface: downstream consumers use standardized, classified data without knowing the original source structure.
SAF-P08	Data is shared property	Standardized data, once registered, is available to all authorized consumers. A source is standardized once and then reused by every product that needs it. Standardizing the same source per product wastes effort and leads to inconsistent coding.

Principle	Statement	Application in VS-02 Data Standardization and Integration
SAF-P09	No data without metadata and classification	VS-02 Data Standardization and Integration is the main point where this principle is fulfilled. Every standardized dataset carries its mappings, classification versions, harmonized definitions, recorded units and periods, and a classification for availability, integrity, and confidentiality. The principle does not require raw data to arrive classified; it requires that data handed over from State #2 onward carries this metadata.
SAF-P10	Security by design	Access controls, integrity verification, and audit logging are built into the pipeline, so every mapping and coercion operation can be traced and tampering can be detected.
SAF-P11	Privacy by design	Direct identifiers are removed or pseudonymized during standardization, at the latest. Data minimization and purpose limitation are applied by design, and pseudonymization keys are kept separately from the data.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 is the defining principle of VS-02: Design defines the target classifications and mappings, Execution applies and records them, and Management assesses classification quality and closes remaining gaps. SAF-P11 applies to *Design Standardization Registration and Privacy Framework*, to *Build Conformance Validation Pipelines and Metadata Registration*, and to the stages VS-02.4 and VS-02.5. SAF-P07 governs both the intake of State #1 from VS-01 and the handover of State #2 to VS-03.

8. Interfaces, services, and events

Boundary interfaces

VS-02 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-01 Data Acquisition and Ingestion → VS-02 Data Standardization and Integration (intake)	Registered raw data version (#1)	Minimum metadata set of State #1: provenance, source, collection method, receipt timestamp, file hash, agreement reference, quality assessment, classification	*Raw Data Registered*	VS-01 Data Acquisition and Ingestion keeps custody of the raw artifact; VS-02 Data Standardization and Integration takes responsibility for standardization on receipt of the event
VS-02 Data Standardization and Integration → VS-03 Data Editing and Enrichment (handover)	Registered standardized data version (#2)	Minimum metadata set of State #2, including the list of unmapped values, the recorded units, currencies, and reference periods, the definition-difference records, and the pseudonymization record	*Standardized Data Registered*	VS-02 Data Standardization and Integration keeps custody of the standardized artifact; VS-03 Data Editing and Enrichment takes responsibility for editing, interpretive coding, conversion, and reconciliation on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Standardization service	VS-01 Data Acquisition and Ingestion (as source), standardization teams	VS-02.1 Profile and Assess Data Structure; VS-02.2 Apply Standard Classifications and Formats	Event subscription (*Raw Data Registered*)
Classification mapping service	Standardization teams, methodologists, other domains that need code mapping	VS-02.2 Apply Standard Classifications and Formats	Mapping interface (source code to standard code), SDMX code list exchange, correspondence table lookup
Semantic harmonization service	Metadata architects, downstream analysts	VS-02.3 Align Metadata and Semantics	Concept system and metadata catalog
Conformance validation service	Process owner, quality manager	VS-02.4 Validate Structural and Semantic Conformance	Conformance reports and dashboards
Standardized data provisioning service	VS-03 Data Editing and Enrichment, other products reusing the source	VS-02.5 Register Standardized Artifacts	Event notification (*Standardized Data Registered*) and a catalog query interface to find standardized datasets and their lineage

Service	Offered to	Realized in	Interfaces
Classification catalog service	All data consumers in the NSI	Management sub-process *Manage Classification and Metadata Lifecycle*	Search and browse interface for classifications, code lists, and concepts, with metadata exchange compliant with SDMX
Design and methodology service	Methodologists, data architects, classification stewards	All Design sub-processes of VS-02 Data Standardization and Integration	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Raw Data Registered	Start of VS-02.1 Profile and Assess Data Structure for that raw version (published by VS-01 Data Acquisition and Ingestion)
Classification version changed	Review of correspondence tables; possible restandardization of existing #1 versions
Standardization Conformed / *Standardization Non-Conformant*	Outcome of VS-02.4 Validate Structural and Semantic Conformance; non-conformance starts re-mapping or conditional acceptance
Standardized Data Registered	Completion of VS-02.5 Register Standardized Artifacts; triggers VS-03 Data Editing and Enrichment
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-02 event choreography between VS-01, the standardization teams, and process management across Design, Implementation, Execution (profile, map and coerce, validate, conformed or non-conformant, register), and Management (key performance indicators, mapping quality, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-02	Receives from VS-02
VS-07 Governance and Compliance	Privacy policy, classification and standards policy	Audit trails, privacy and standards compliance evidence
VS-08 Innovation and Methodological Development	New mapping methods, support for coding	Recurring mapping problems, improvement needs
VS-09 Infrastructure and Platform Enablement	Mapping and classification services, storage, catalog, event interfaces	Capacity and service requirements
VS-10 Customer and Stakeholder Engagement	Requirements of users for classifications and comparability	Information on the classifications and versions applied

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Mapping coverage	Share of source codes mapped one-to-one to a valid standard code	Per dataset, per classification
Unmapped-value rate	Share of values without a correspondence, routed to VS-03 Data Editing and Enrichment	Per source, per cycle
Conformance pass rate	Share of datasets that pass the gate of State #2 on the first run	Per source, per cycle
Format conformance rate	Share of records that conform to the target structure and format	Per dataset, per cycle
Processing time	Time from the *Raw Data Registered* event to the registration of #2	Per dataset, per cycle
Re-mapping rate	Share of datasets returned for re-mapping or design change	Per source, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-02 are part of the SAF Toolkit.

Toolkit: SAF-TK-V02 VS-02 Data Standardization and Integration companion [**VC17**]